Q3 Solutions

Consider the numerical variables in the data set and find the best SIMPLE linear regression model to predict the prices. Test the assumptions and use transformations if it is required. Explain why the model you find is the best simple linear regression model and interpret the model coefficients.

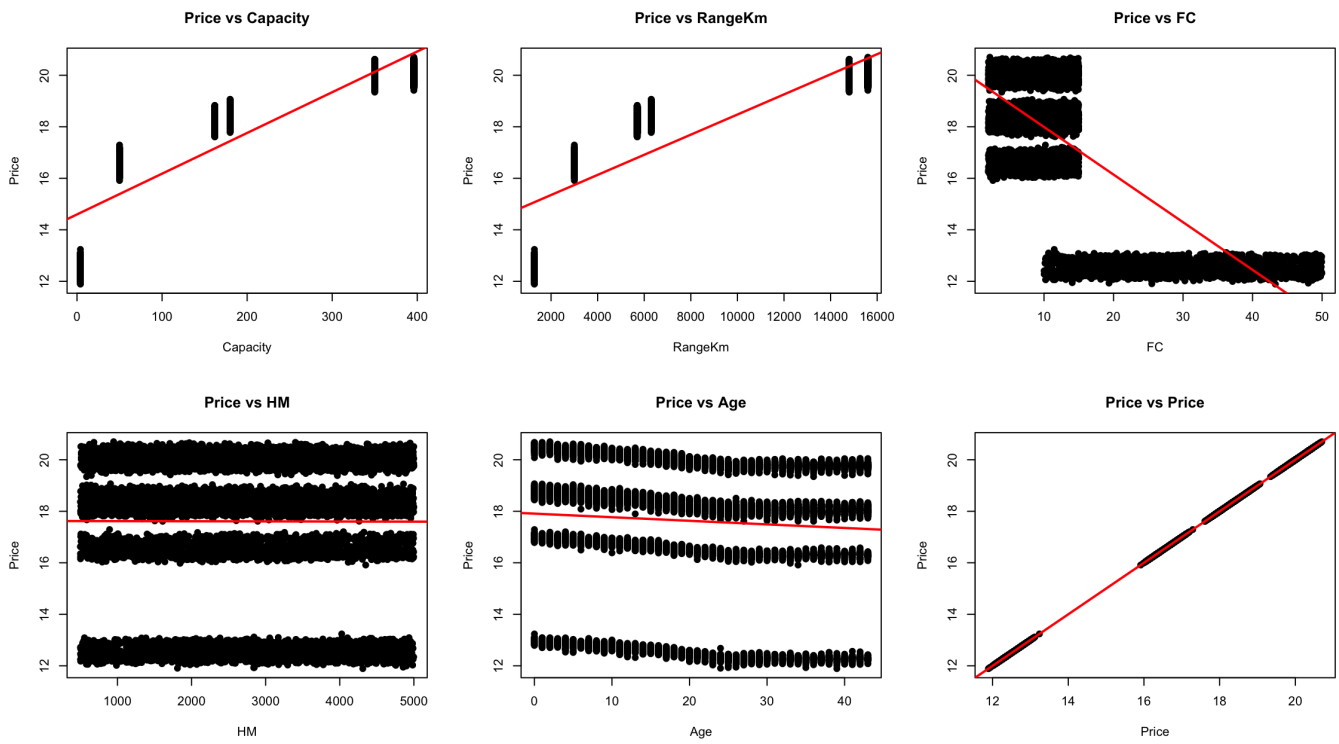

Figure 01

According to plot analysis, we selected the variables `Capacity`, `RangeKm`, and `FC` (Fuel Consumption) as the best candidates for the linear model because they show the most linear relationship with `Price`. A `log()` transformation was applied to `Price` to normalize its distribution.

The correlation matrix confirmed that `Capacity`, `RangeKm`, and `FuelConsumption` are the top 3 numerical variables with the highest correlation to `Price`.

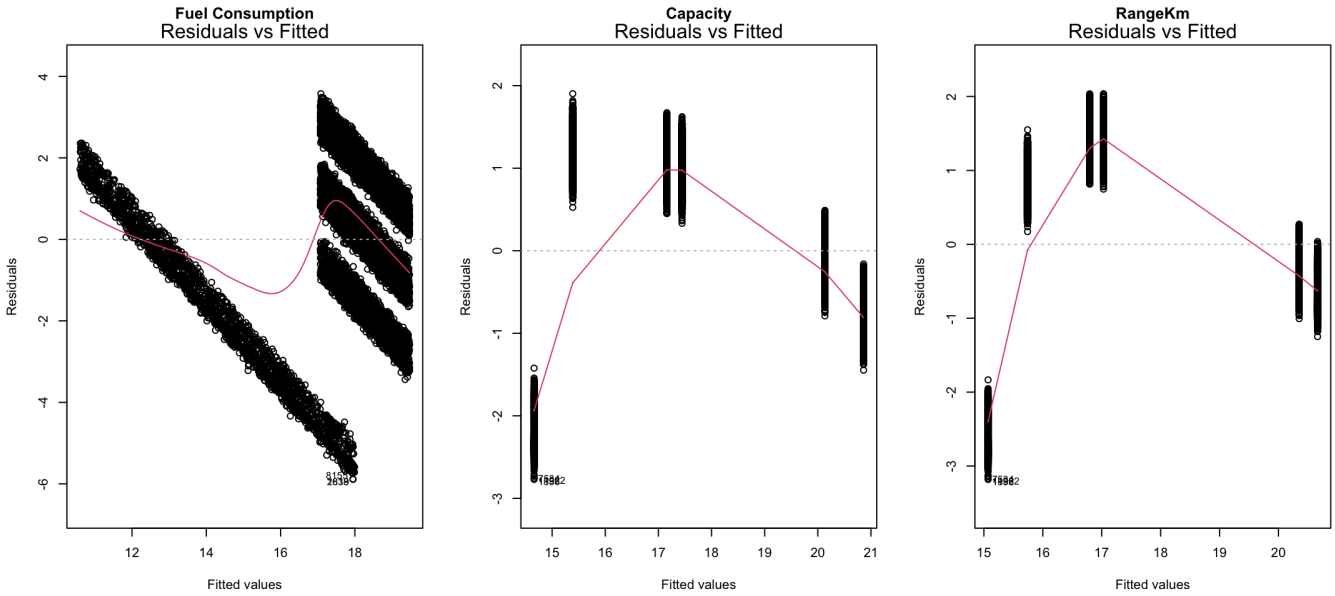

Figure 02

The Residual vs Fitted plots suggested that the model may need a quadratic term. After comparing `Capacity` vs `RangeKm` using a quadratic term, the ANOVA shows that `RangeKm` + `RangeKm^2` yields the lowest RSS results.

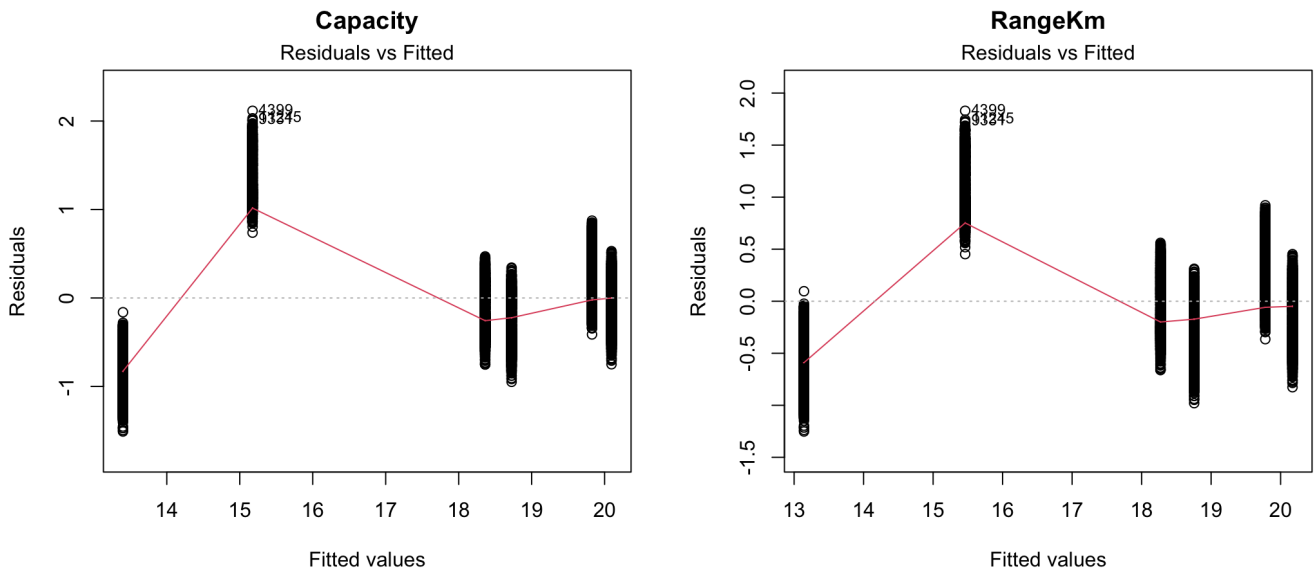

Figure 2.1

The new model with the quadratic term shows better residuals.

Regression assumptions analysis

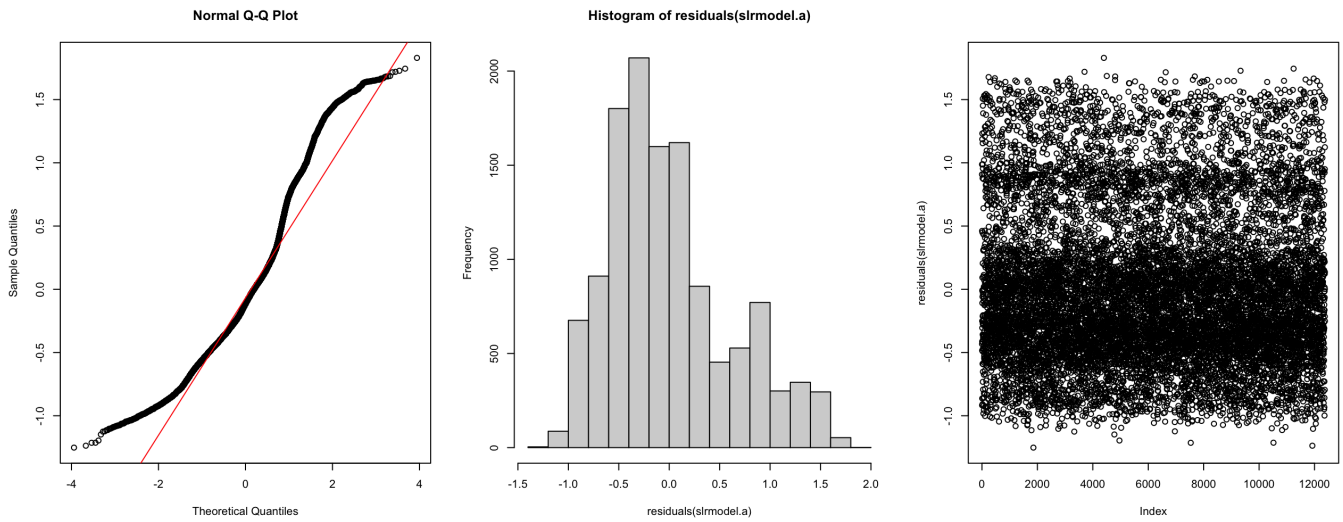

Figure 03

> This model shows p-value < 0.05 on the Breusch-Pagan, meaning it has signs of heteroscedasticity.

Conclusion

```
Call:
lm(formula = Price ~ RangeKm + RangeKm_2, data = numeric_air_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.2519 -0.4347 -0.1157  0.2976  1.8296 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  1.113e+01  1.752e-02   635.5   <2e-16 ***
RangeKm      1.654e-03  5.508e-06   300.4   <2e-16 ***
RangeKm_2   -7.052e-08  3.023e-10  -233.3   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6079 on 12374 degrees of freedom
Multiple R-squared:  0.944,	Adjusted R-squared:  0.944 
F-statistic: 1.043e+05 on 2 and 12374 DF,  p-value: < 2.2e-16
```

Figure 04

We selected `RangeKm` (along with its quadratic transformation) as the best single predictor because it shows a high correlation with Price and explains the highest proportion of the variance in airplane prices.

- **Intercept (β_0)**: The expected log(Price) or baseline price of a plane.
- **Slope for RangeKm (β_1 & β_2)**: For each additional `RangeKm`, the log(Price) increases by β_1 and β_2, meaning Price increases exponentially by e^{β_2}

Fit a multivariate linear regression model with the most important two (numerical) variables. Use transformations if it is needed and test all the assumptions. Then compare this model to the simple linear regression model that you fit in (a). Which one is a better model? Why?

```
regmodel.all <- lm(Price ~ ., data = numeric_air_data)
summary(regmodel.all)
```

```
Call:
lm(formula = Price ~ ., data = numeric_air_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.8521 -0.3464  0.1172  0.5222  1.6157 

Coefficients:
              Estimate Std. Error  t value Pr(>|t|)    
(Intercept)  1.682e+01  2.569e-02  654.989   <2e-16 ***
Capacity     3.080e-02  3.505e-04   87.856   <2e-16 ***
RangeKm     -4.671e-04  8.884e-06  -52.576   <2e-16 ***
FC          -8.760e-02  8.055e-04 -108.751   <2e-16 ***
HM           2.249e-06  5.190e-06    0.433    0.665    
Age         -1.819e-02  5.258e-04  -34.590   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.7465 on 12371 degrees of freedom
Multiple R-squared:  0.9156,	Adjusted R-squared:  0.9155 
F-statistic: 2.683e+04 on 5 and 12371 DF,  p-value: < 2.2e-16
```

Figure 05

First, we checked a model against all our top variables: `Capacity`, `RangeKm`, `FC`, and `Age`.

```
regmodel.best4 <- lm(Price ~ Capacity + RangeKm + FC + Age, data =
numeric_air_data)
summary(regmodel.best4)

anova(regmodel.all, regmodel.best4)
```

The ANOVA confirmed that the model with all predictors is not significantly better than the model with these 4 predictors. However, checking Variance Inflation Factors (VIF) revealed critical multicollinearity.

```
cor(numeric_air_data[,-price.col])
vif(regmodel.best4)
```

```
> cor(numeric_air_data[,-price.col])
            Capacity     RangeKm          FC          HM         Age
Capacity  1.00000000  0.98950931 -0.46818683 -0.01187781  0.01920662
RangeKm   0.98950931  1.00000000 -0.42415129 -0.01369395  0.01855716
FC       -0.46818683 -0.42415129  1.00000000 -0.01123132 -0.02029864
HM       -0.01187781 -0.01369395 -0.01123132  1.00000000  0.01047742
Age       0.01920662  0.01855716 -0.02029864  0.01047742  1.00000000
> vif(regmodel.best4)
 Capacity   RangeKm        FC       Age
55.540733 52.879523  1.413722  1.000534
```

Figure 06

The Correlation Matrix and VIF tell us that Capacity and RangeKm are highly correlated. VIF also tell us that there is a multicollinearity problem within our predictors. We choose to drop the predictor with the highest VIF: `Capacity`

Now, VIF shows no multicollinearity among the selected predictors.

By systematically dropping predictors and comparing via ANOVA, we discovered that keeping `RangeKm` and `FC` led to higher R^2 and a smaller Sum of Square Residuals. Similar to the SLR, adding a quadratic term `RangeKm^2` further improved the model.

Model A vs Model B comparison

```
> anova(slrmodel.a, regmodel.b)
Analysis of Variance Table

Model 1: Price ~ RangeKm + RangeKm_2
Model 2: Price ~ RangeKm_2 + RangeKm + FC
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1  12374 4572.8
2  12373 3321.5  1    1251.3 4661.3 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 07

- The MLR model with RangeKm + RangeKm^2 + FC has a higher adjusted R-squared and a lower residual standard error than the SLR with Capacity alone, indicating a better fit.
- The ANOVA F-test confirms that adding the second variable significantly improves the model.

Regression assumptions analysis

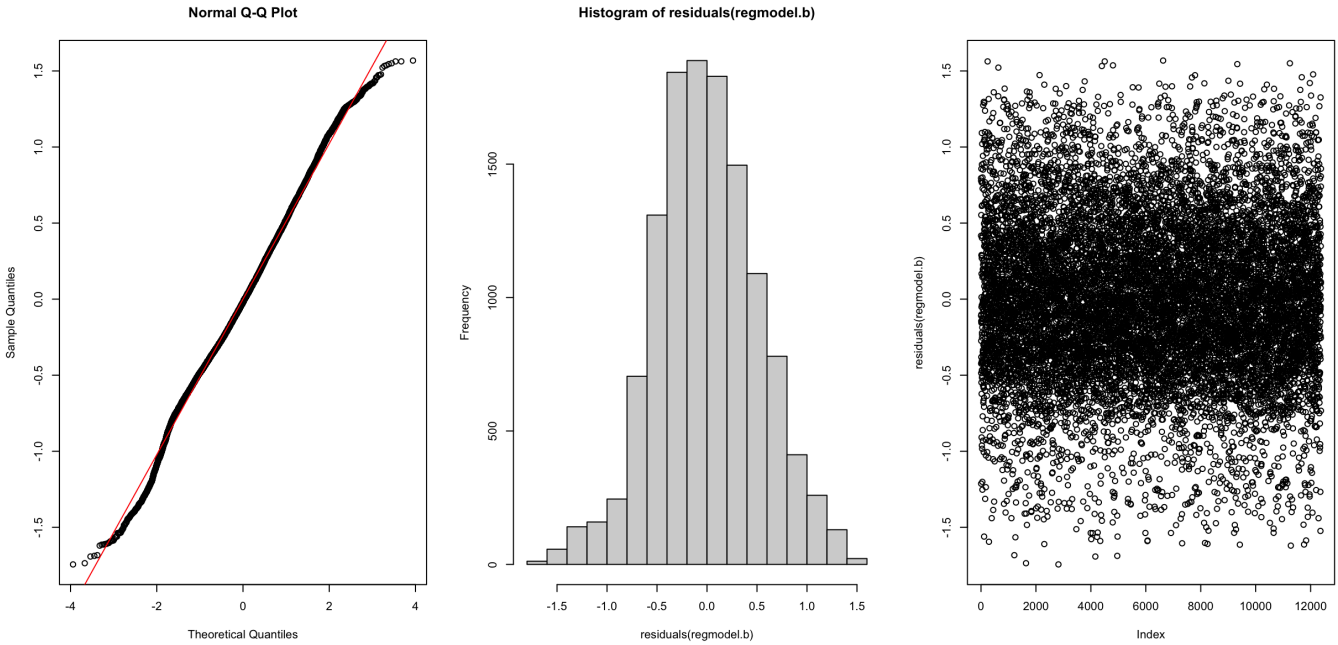

Figure 08

Encode the variable model into three categories considering whether the plane is “Airbus”, “Boeing” or “Other”. Now add this model factor to the regression model you have chosen in section (b). Interpret the coefficients and overall summary of the model. Compare the model in section (b) with the model that has an additional factor. Which one would you choose? Why?

We added ModelCat to our selected model. However, introducing `ModelCat` made `FC` statistically insignificant. We dropped `FC` to simplify the model, yielding: `lm(Price ~ RangeKm_2 * ModelCat + RangeKm)`.

```
Call:
lm(formula = Price ~ RangeKm_2 * ModelCat + RangeKm, data = numeric_air_data)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.62176 -0.21716 -0.07364  0.22958  0.75755 

Coefficients: (1 not defined because of singularities)
                           Estimate Std. Error t value Pr(>|t|)    
(Intercept)               5.233e+00  2.792e-02  187.46   <2e-16 ***
RangeKm_2                -1.288e-07  2.970e-10 -433.75   <2e-16 ***
ModelCatBoeing            5.520e-01  1.009e-02   54.69   <2e-16 ***
ModelCatOther             3.755e+00  1.741e-02  215.72   <2e-16 ***
RangeKm                   2.901e-03  6.150e-06  471.76   <2e-16 ***
RangeKm_2:ModelCatBoeing  1.332e-09  6.122e-11   21.76   <2e-16 ***
RangeKm_2:ModelCatOther          NA         NA      NA       NA    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2686 on 12371 degrees of freedom
Multiple R-squared:  0.9891,	Adjusted R-squared:  0.9891 
F-statistic: 2.24e+05 on 5 and 12371 DF,  p-value: < 2.2e-16
```

Figure 09

Conclusion

- Intercept (β_0): Expected log(Price) for the baseline category.
- ModelCat effects: The coefficients for Boeing and "Other" represent the shift in the baseline log(Price) compared to Airbus.
- Interaction (RangeKm_2 * ModelCat): The slope for the squared range changes depending on the airplane manufacturer.

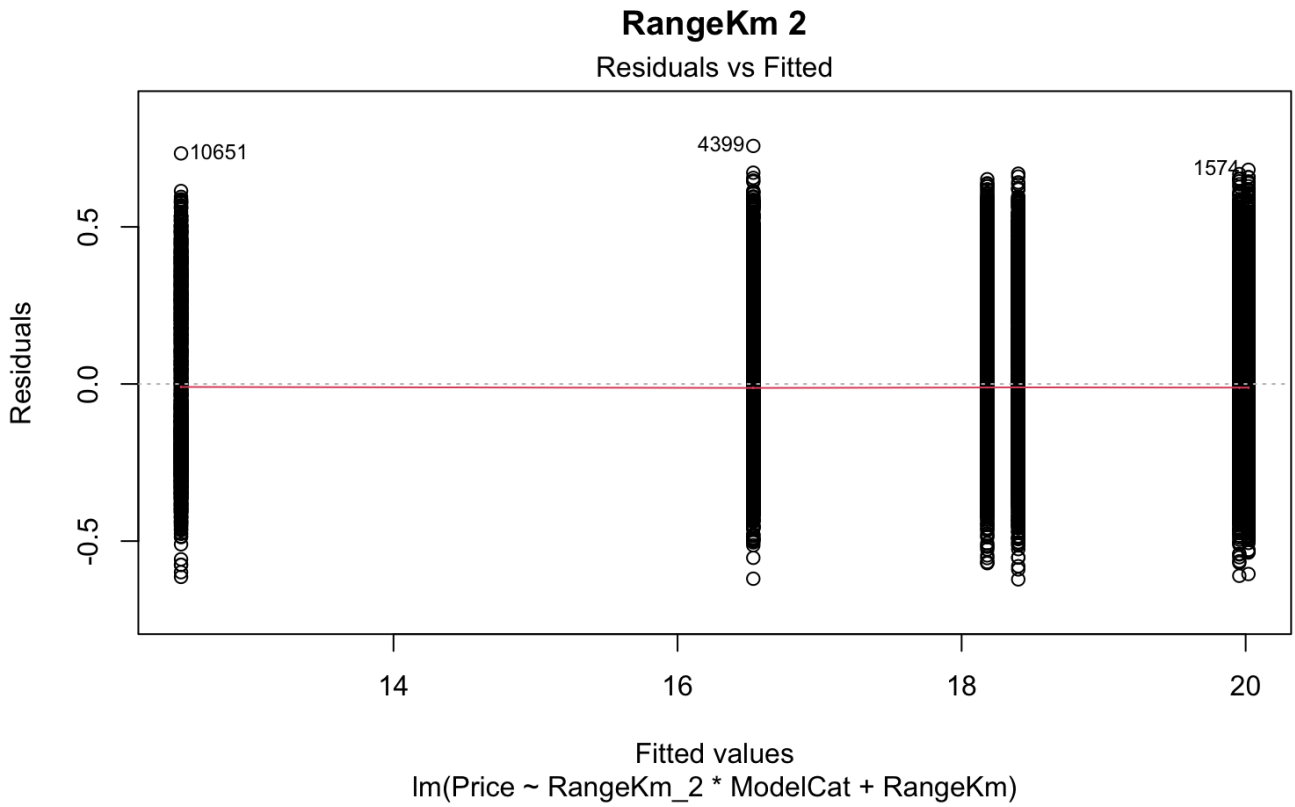

Figure 10

The Residual vs Fitted plot shows good results.

Model B vs Model C comparison

```
> anova(regmodel.b_2, regmodel.c_2)
Analysis of Variance Table

Model 1: Price ~ RangeKm_2 + RangeKm + FC
Model 2: Price ~ RangeKm_2 * ModelCat + RangeKm + FC
  Res.Df    RSS Df Sum of Sq     F    Pr(>F)
1  12373 3321.5
2  12370  892.2  3    2429.3 11227 < 2.2e-16 ***
---
```

Figure 11

- The ANOVA analysis tells us that adding ModelCat decreases the Residual Sum of Squares, meaning the model significantly improves with this new variable.
- We selected this interaction model (Model C) as our best model.

Test the validity of the final model that you choose.

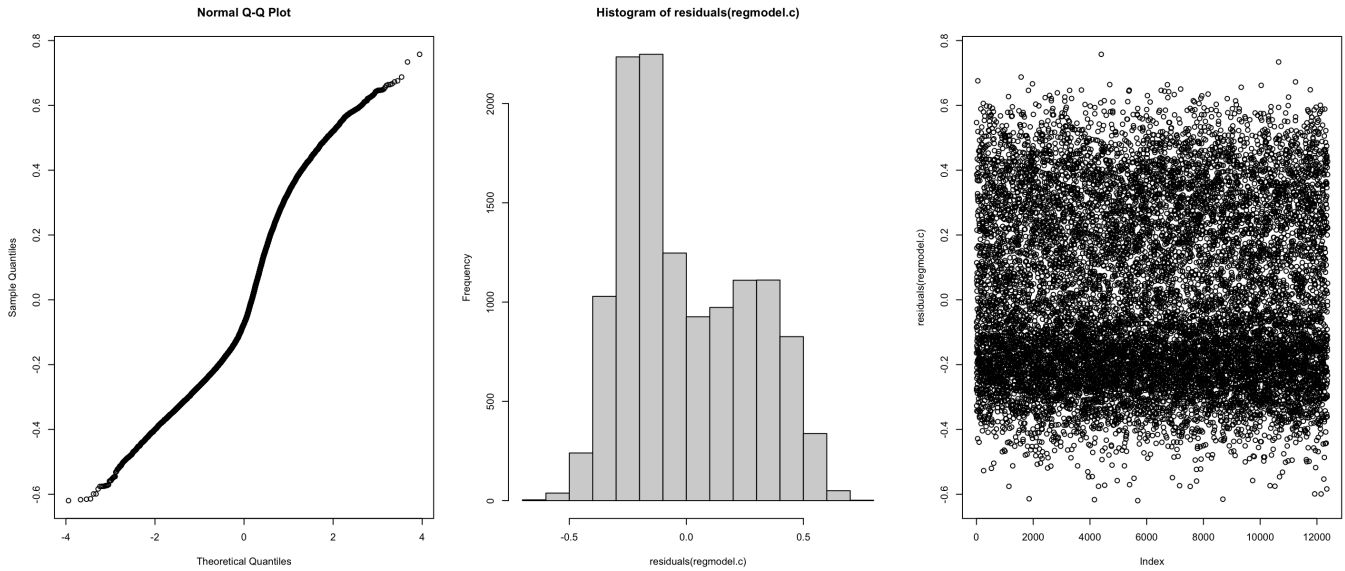

Figure 12

To fully validate the predictive power of our chosen model `(Price ~ RangeKm_2 * ModelCat + RangeKm)`, we split the data into an 80% training set and a 20% testing set.

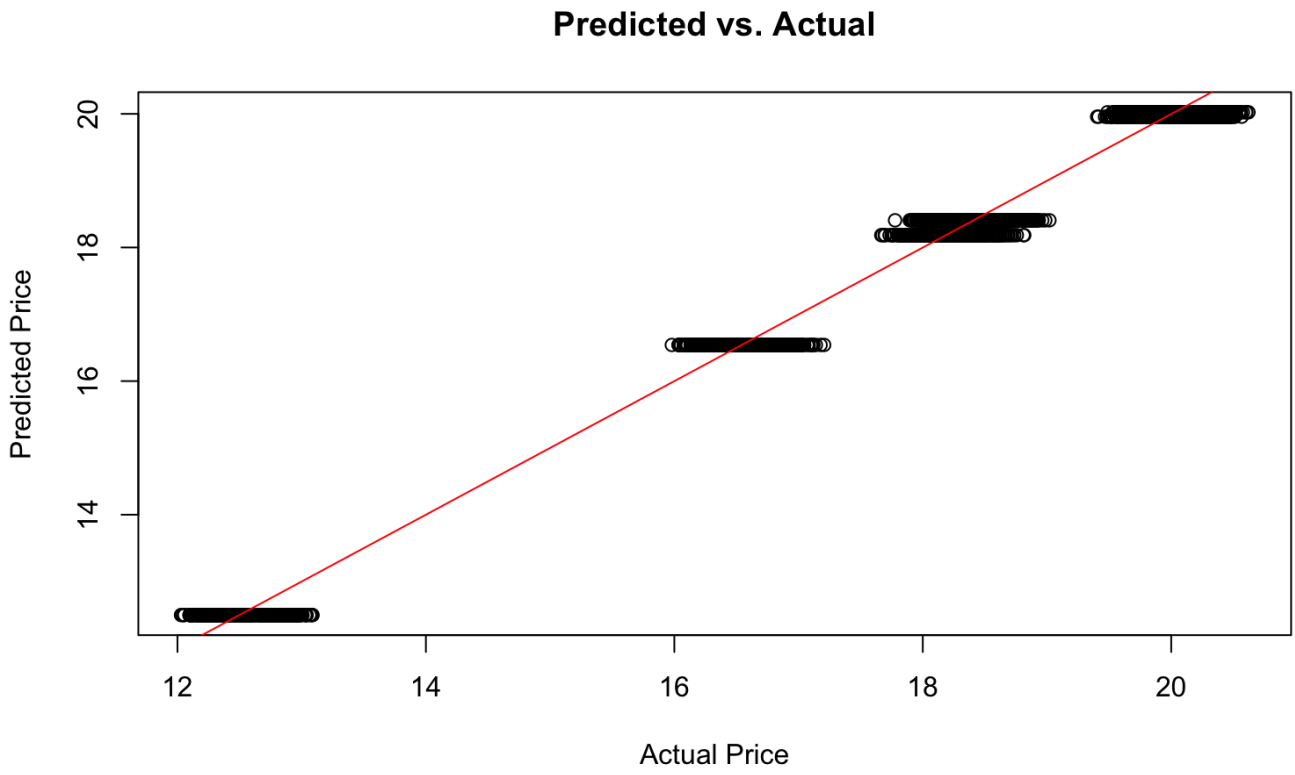

Figure 13

Validity and Performance

- Out-of-sample MSE: We calculated the Mean Squared Error against the holdout Test Set, which quantifies the average squared prediction error on unseen data.
- Diagnostics: The Q-Q plot of the final model mostly follows the diagonal (normality holds). Durbin-Watson statistic is close to 2 (no autocorrelation).

- Plotting `Actual vs Predicted` values from the test set showed data points clustering closely around the ideal `y = x` reference line, meaning the model generalizes well and confirms our findings.